

LEAP 2025 Mathematics

Practice Test

Grade 6

Session 1

Directions:

Today, you will take Session 1 of the Grade 6 Mathematics Practice Test. You will not be able to use a calculator in this session.

Read each question. Then, follow the directions to answer each question. Mark your answers by circling the correct choice. If you need to change an answer, be sure to erase your first answer completely.

Some of the questions will ask you to write a response. Write your response in the space provided in your test booklet.

If you do not know the answer to a question, you may go on to the next question. If you finish early, you may review your answers and any questions you did not answer in this session **ONLY**.

GO ON ►

1. A can contains $\frac{15}{16}$ pound of vegetables. One serving of these vegetables weighs $\frac{1}{4}$ pound.

What is the total number of servings of vegetables in the can?

- A. $\frac{15}{64}$ serving
- B. $\frac{4}{15}$ serving
- C. $1\frac{3}{16}$ servings
- D. $3\frac{3}{4}$ servings

2. Which of these expressions are equivalent to $\frac{p}{3}$?

Select **each** correct answer.

A. $p - \frac{2}{3}p$

B. $\frac{1}{3}p$

C. $p - 3$

D. $3 \div p$

E. $\frac{3p}{9}$

F. $\frac{1}{3}p + \frac{1}{3}p + \frac{1}{3}p$

3. This table shows the numbers of books, by type, checked out from the school library on Monday.

Book Checkout

Book Type	Number of Books
mystery	24
nonfiction	18
adventure	12
humor	16

Select a number from each list to complete the statement.

For every _____ mystery books checked out,

2
3
4
6

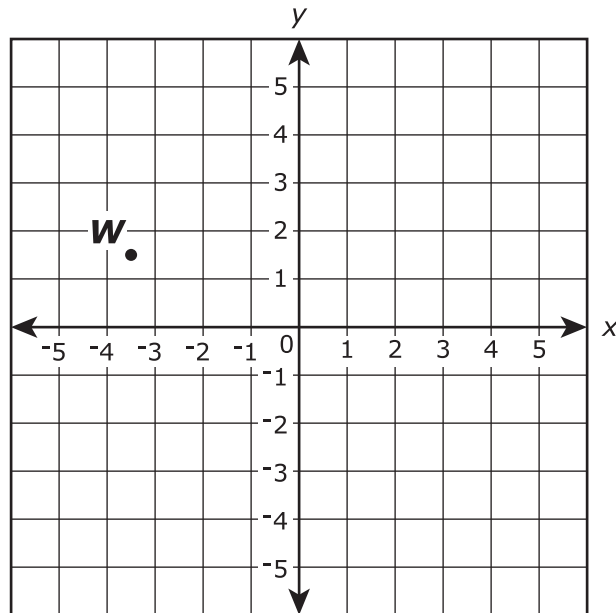
_____ nonfiction books were checked out.

2
3
4
6

GO ON ►

4. Sonia has two packages of hamburger meat. The first package weighs 1.76 pounds and the second package weighs 2.29 pounds. She mixes the two packages together and forms hamburgers that weigh 0.25 pound each. What is the greatest number of 0.25-pound hamburgers Sonia can make using the hamburger meat she has?
- A. 2
 - B. 7
 - C. 9
 - D. 16

5. This coordinate plane shows the location of point W .



What is the value of the x -coordinate of point W ? Enter your answer as a decimal to the nearest 0.5.

Enter your answer in the box.

6. Enter your answer in the box.

$$34,992 \div 81 =$$

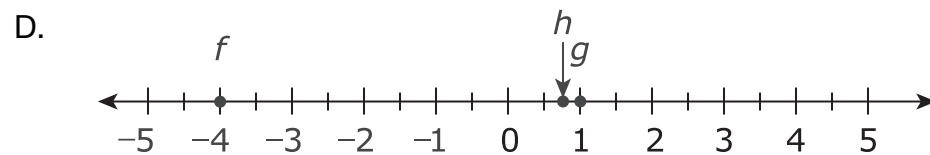
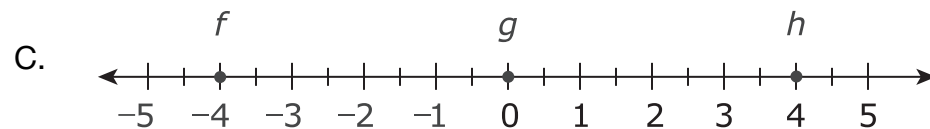
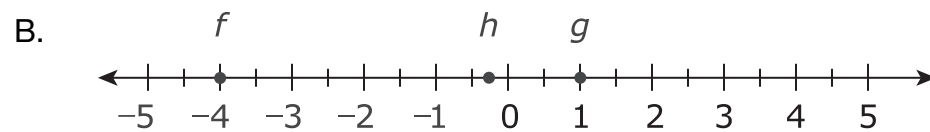
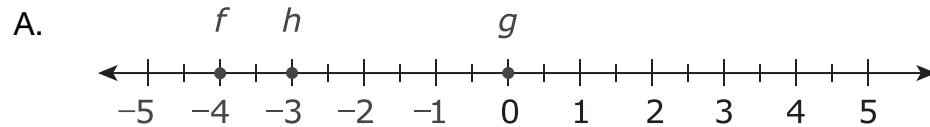
7. Three values on a number line are labeled f , g , and h .

$$f = -4$$

$$g = -g$$

$$h = -f$$

Which number line correctly shows the values of f , g , and h ?



GO ON ►

8. Select the expression that is equivalent to $48 + 12$.

- A. $6(8 + 6)$
- B. $12(4 + 1)$
- C. $4(44 + 3)$
- D. $8(6 + 4)$

9. Which questions are statistical questions?

Select **each** correct answer.

- A. How old is Mr. Patterson?
- B. How many states has Juanita visited?
- C. How many students are in Mrs. Lee's class today?
- D. How many students eat lunch in the cafeteria each day?
- E. How many pets does each student at your school have at home?

GO ON ►

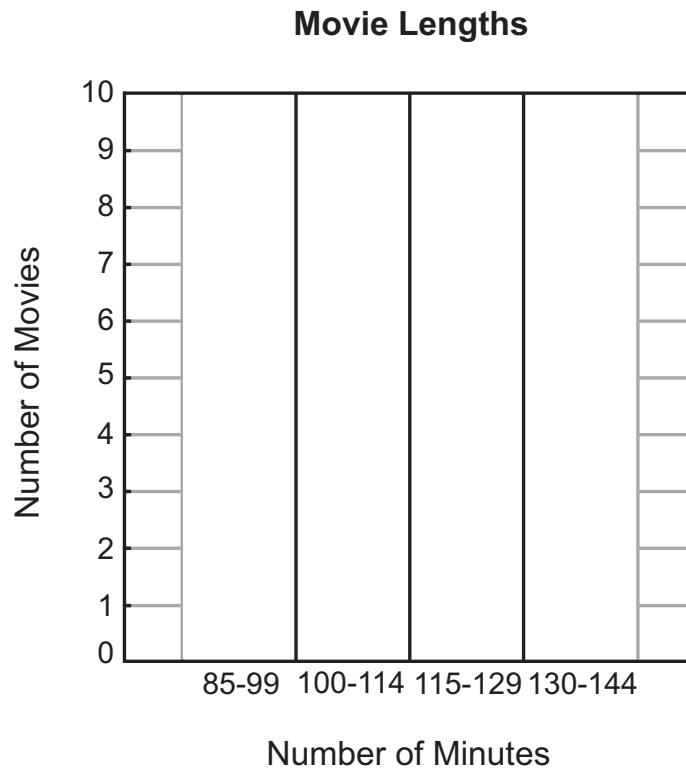
10. Sally rents a life jacket for a one-time fee of \$5. She then rents a canoe for \$15 per hour. Which expression represents the total cost, in dollars, to rent the life jacket and the canoe for h hours?
- A. $5 + 15h$
 - B. $10h$
 - C. $15 + 5h$
 - D. $20h$

11. The lengths, in minutes, of the movies currently showing at a movie theater are shown in the data set.

89	98	109	123	123	125	125
128	130	135	137	140	143	143

Create a histogram that represents the data.

Draw each bar to the correct height.



12. Water freezes at 0° Celsius. The table shows five different temperatures in degrees Celsius. Indicate whether each temperature is above or below freezing.

Mark an X in one cell per row.

	Above Freezing	Below Freezing
0.5° C		
-13° C		
100° C		
5.5° C		
-2.25° C		

- 13.** What is the result when 75,069 is divided by 45?

Enter your answer in the box.

14. An expression is shown.

$$12 \cdot 12 \cdot 12 \cdot 12 + 7(3 \cdot 3 \cdot 3 \cdot 3 + 3)$$

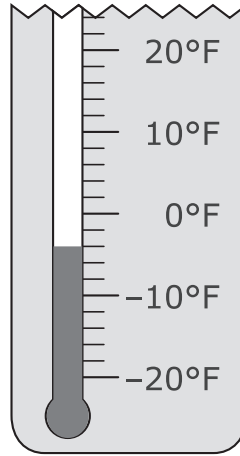
Which of the following shows this expression written using exponents?

- A. $4^{12} + 7(5^3)$
- B. $4^{12} + 7(4^3 + 3)$
- C. $12^4 + 7(3^5)$
- D. $12^4 + 7(3^4 + 3)$

- 15.** Heather drove at a constant rate. She traveled 162 miles in 3 hours. How far, in miles, did Heather travel in 1 hour?

Enter your answer in the box.

16. The picture shows part of a thermometer measuring temperature in degrees Fahrenheit.



What is the temperature, in degrees Fahrenheit, shown on the thermometer to the nearest integer?

Enter your integer answer in the box.

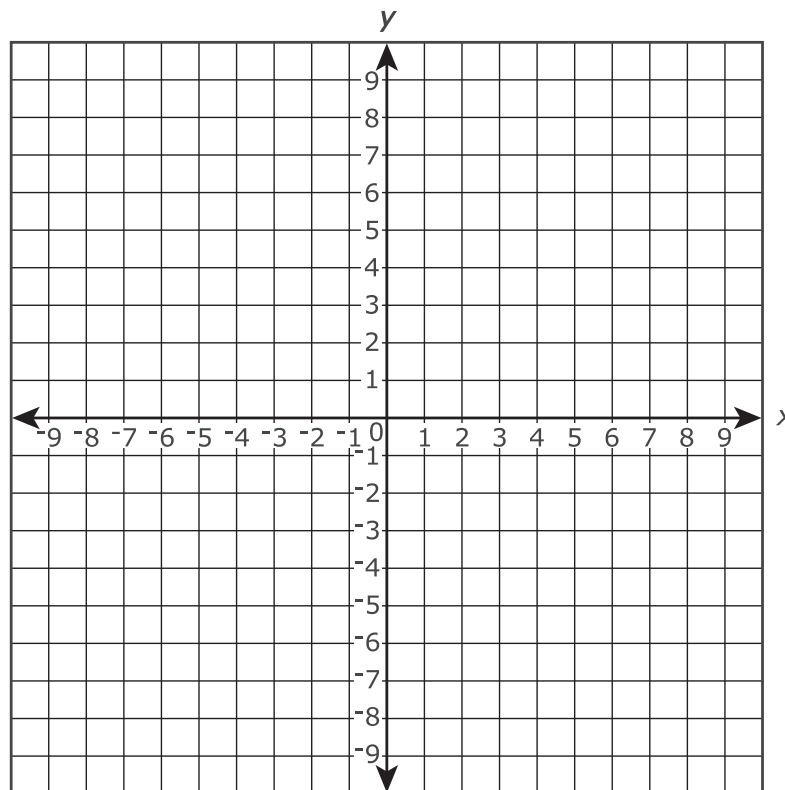
17. City planners are creating a neighborhood map on a coordinate grid. The table shows the locations of the neighborhood library and school on a coordinate grid.

Neighborhood Planning

Building	Location
library	$(-4, -6)$
school	$(5, -6)$

In this coordinate grid, the distance between each gridline represents 1 mile. What is the distance, in miles, between the library and the school?

You can use the coordinate grid to help you find the answer by plotting the two points. Be sure to place your final answer in the box.



Enter your answer in the box.

GO ON ►

18. Select each expression that is equivalent to $3(n + 6)$.

Select **all** that apply.

- A. $3n + 6$
- B. $3n + 18$
- C. $2n + 2 + n + 4$
- D. $2(n + 6) + (n + 6)$
- E. $2(n + 6) + n$

19. What is $78.32 + 6.784$?

Enter your answer in the box.

GO ON ►

20. What is the quotient of $33.32 \div 9.8$?

- A. 2.9
- B. 3.4
- C. 3.6
- D. 4.1



Session 2

Directions:

Today, you will take Session 2 of the Grade 6 Mathematics Practice Test. You will be able to use a calculator in this session.

Read each question. Then, follow the directions to answer each question. Mark your answers by circling the correct choice. If you need to change an answer, be sure to erase your first answer completely.

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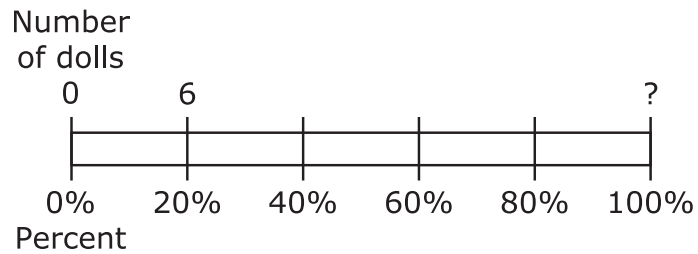
GO ON ►

21. Which verbal expression best represents the algebraic expression $3n - 4$?

- A. three less than four times a number
- B. four less than three times a number
- C. three times four subtracted from a number
- D. four times the difference of a number and three

GO ON ►

22. Anita brings 6 dolls to her grandma's house. These dolls represent 20% of Anita's doll collection, as shown in the diagram.



What is the total number of dolls in Anita's doll collection?

Enter your answer in the box.

- 23.** Hank bought 5 meters of ribbon for \$4.

Select from each list to complete the sentence.

The ribbon costs _____ per _____.

\$0.008
\$0.08
\$0.80

millimeter
centimeter
kilometer

- 24.** Let x represent any number in the set of even integers greater than 1.

Which inequality is true for all values of x ?

- A. $x < 0$
- B. $x > 0$
- C. $x < 4$
- D. $x > 4$

- 25.** Evaluate the expression $120n + 160,500$ when $n = 32$.

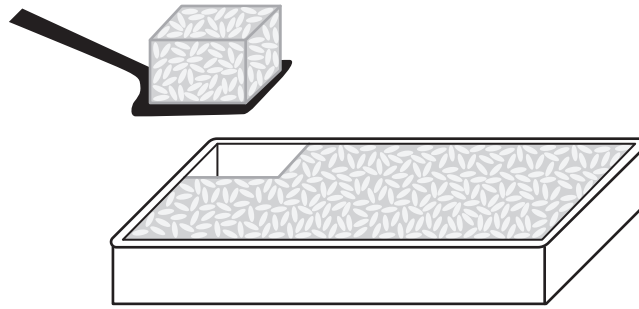
Enter your answer in the box.

26. What is the value of 0.2^3 ?

Enter your answer in the box.

GO ON ►

27. Megan spent \$9.85 on ingredients and made one pan of cereal bars. The pan has a length of 24 inches and a width of 16 inches.



Megan needs to cut individual cereal bars from the pan. Each cereal bar should be the same size and shape and should represent a reasonable serving.

Estimate an appropriate length and width for each cereal bar and explain your assumptions.

Based on your estimate, determine the amount each cereal bar will cost Megan to make. Show your work or explain your reasoning.

Enter your answers and your work or explanations in the box provided.

GO ON ►

28. The number of blueberry muffins that a baker makes each day is 40% of the total number of muffins she makes.

Part A

On Monday, the baker makes 36 blueberry muffins.

What is the total number of muffins that the baker makes on Monday?

Enter your answer in the box.

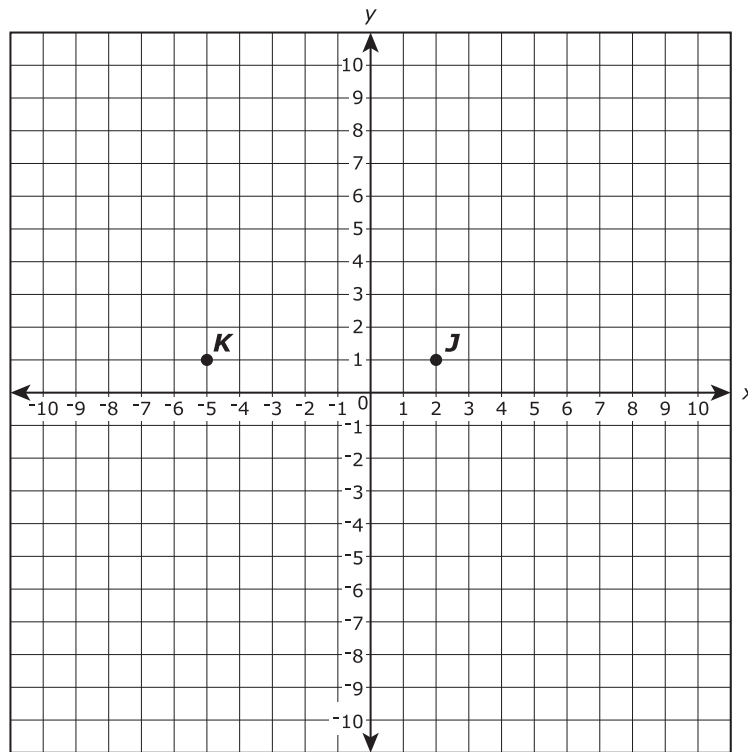
Part B

On Tuesday, the baker makes a total of 60 muffins.

How many blueberry muffins does the baker make on Tuesday?

Enter your answer in the box.

29. Points J and K , plotted on the coordinate grid, are two vertices of rectangle $JKLM$. Rectangle $JKLM$ has an area of 7 square units. Point J is located at $(2, 1)$, and point K is located at $(-5, 1)$. Each vertex of the rectangle is located at a point that has integer coordinates.



Part A

Which points could be another vertex of the rectangle?

Select **each** correct answer.

- A. $(-5, 0)$
- B. $(-5, 2)$
- C. $(1, 1)$
- D. $(2, -6)$
- E. $(2, 0)$
- F. $(9, 1)$
- G. $(2, 2)$

Part B

What is the perimeter of rectangle $JKLM$?

Enter your answer in the box.

- 30.** A student made two patterns to show multiplication of a decimal by powers of ten. The equations shown for both patterns are incorrect.

Pattern A

$$3.675 \bullet 10 = 3.6750$$

$$3.675 \bullet 100 = 3.67500$$

$$3.675 \bullet 1,000 = 3.675000$$

Pattern B

$$3.675 \bullet 0.1 = 3.0675$$

$$3.675 \bullet 0.01 = 3.00675$$

$$3.675 \bullet 0.001 = 3.000675$$

Explain why the equations in each of the patterns are false. Include in your explanation the values that should appear on the right side of each equation in both patterns to make the equations true.

Enter your explanation in the box provided.

GO ON ►

31. The students in a club are selling flowerpots to raise money. Each flowerpot sells for \$15.

Part A

Write an expression that represents the total amount of money, in dollars, the students raise from selling x flowerpots.

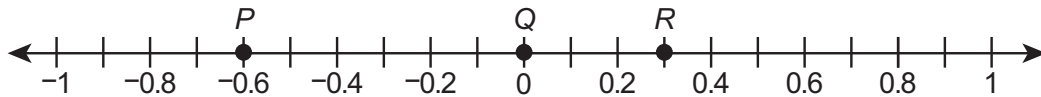
Enter your expression in the box provided. Enter **only** your expression.

Part B

The goal of the students in the club was to raise \$500. They sold 43 flowerpots. By what amount did the students exceed their goal of raising \$500? Show or explain all your work.

Enter your answer and your work or explanation in the box provided.

32. Points P , Q , and R are shown on the number line.

**Part A**

Find the distances between points P and Q and between points R and Q . Show your work or explain your answers. Refer to the number line in your explanation.

Enter your answers and your work or explanation in the box provided.

Part B

Point S is a different point on the number line. Point S and point R are the same distance from point Q . Explain how to determine the location of point S on the number line.

Enter your explanation in the box provided.



Session 3

Directions:

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GO ON ►

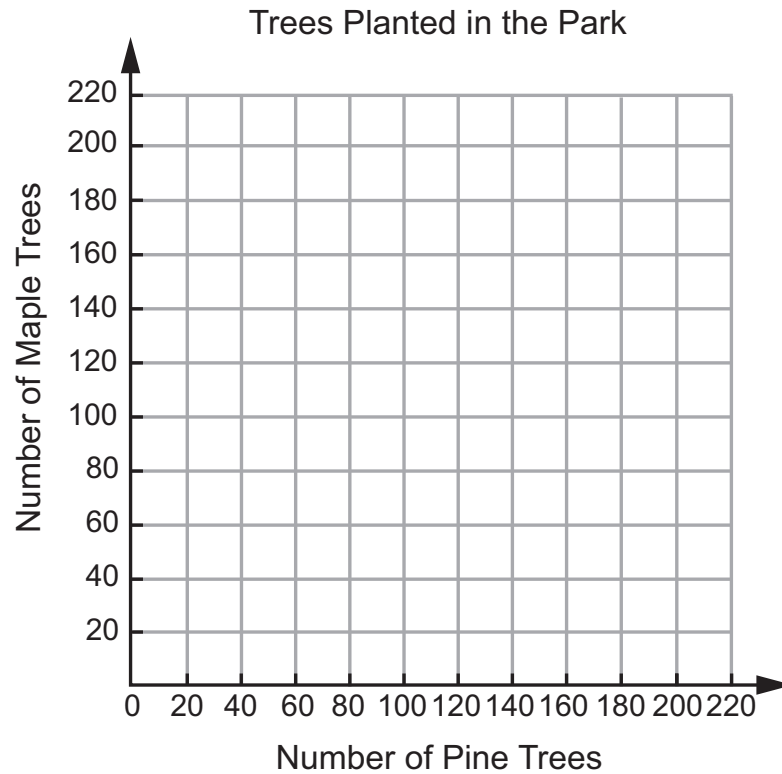
- 33.** Natalie uses a 15% off coupon when she buys a camera. The original price of the camera is \$45.00. How much money, in dollars, does Natalie save by using the coupon?

Enter your answer in the box.

- 34.** A total of 300 trees will be planted in a park. There will be 2 pine trees planted for every 3 maple trees planted.

Plot the point that represents the number of pine trees and the number of maple trees that will be planted.

Select the place on the coordinate plane to plot the point.



- 35.** Jill bought a pound of strawberries for \$4.00. What is the price, in dollars, per **ounce** of strawberries?

Enter your answer in the box.

36. Which expressions represent “the sum of 3 and n ”?

Select **all** that apply.

A. $3n$

B. $n + 3$

C. $3 + n$

D. $n + n + n$

E. n^3

- 37.** Brianna's teacher asks her which of these three expressions are equivalent to each other.

Expression A: $9x - 3x - 4$

Expression B: $12x - 4$

Expression C: $5x + x - 4$

Brianna says that all three expressions are equivalent because the value of each one is -4 when $x = 0$.

Brianna's thinking is incorrect.

- Identify the error in Brianna's thinking.
- Determine which of the three expressions are equivalent.
- Explain or show your process in determining which expressions are equivalent.

Enter your answers and your explanation or process in the box provided.

GO ON ►

38. Kellie bought 8 towels and spent \$39.60. Each towel costs the same amount.

Part A

Select from each list to create an equation that can be used to determine t , the price, in dollars, of 1 towel.

t _____ = _____

+
−
×
÷

8
39.60

8
39.60

Part B

What is the price, in dollars, of 1 towel?

Enter your answer in the box.

--

39. What is the value of $a^2 + 3b \div c - 2d$, when $a = 3$, $b = 8$, $c = 2$, and $d = 5$?

Enter your answer in the box.

GO ON ►



40. Maggie is making a necklace using a 13-inch string and identical beads.

Maggie has placed 12 beads next to each other starting at the left end of the string as shown in the figure. The 12 beads fill 3 inches of the string.

Part A

How many total beads will completely fill the 13-inch string?

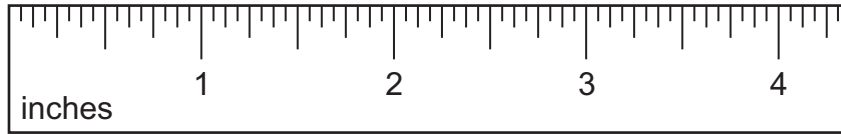
Enter your answer in the box.

Part B

After filling the string from end to end, Maggie decides to leave one inch at each end of the string with no beads. How many beads does she need to remove from the beads she used to completely fill the string?

Enter your answer in the box.

41. One size of cardboard can be purchased in sheets that are $\frac{3}{16}$ inch thick. The sheets of cardboard are stacked on top of each other in packages. The height of each stack is $2\frac{1}{4}$ inches.



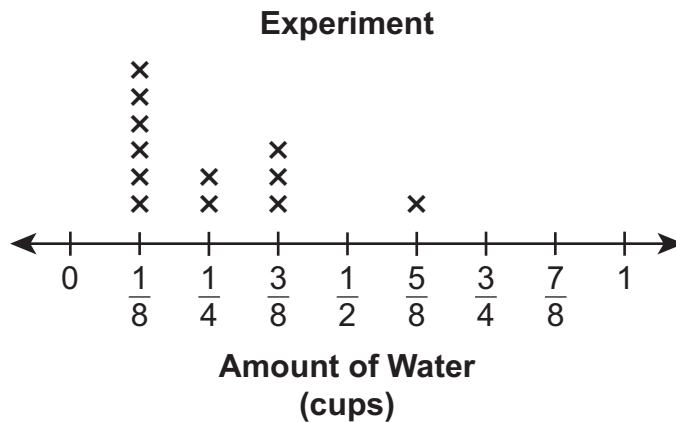
- Use the model of a ruler to determine the number of sheets of cardboard in a stack.
- Explain how you used the model to find your answer.
- Write an expression that can be used to determine the number of sheets of cardboard in a stack.
- Explain how your expression relates to the model.

Enter your answer, your expression, and your explanations in the box provided.

GO ON ►

- 42.** The variable x represents a value in the set $\{4, 6, 7, 8\}$. Which value of x makes $2(x - 4) + 3 = 7$ a true statement?
- A. 4
 - B. 6
 - C. 7
 - D. 8

43. The line plot shows the amount of water used by 12 students during an experiment.



Part A

Write and evaluate an expression using addition and multiplication to determine the total number of cups of water used by the 12 students during the experiment. Show or explain each step you used to evaluate the expression.

Enter your expression and your work or explanation in the box provided.

GO ON ►

Part B

The water used by the 12 students during the experiment was poured from a beaker.

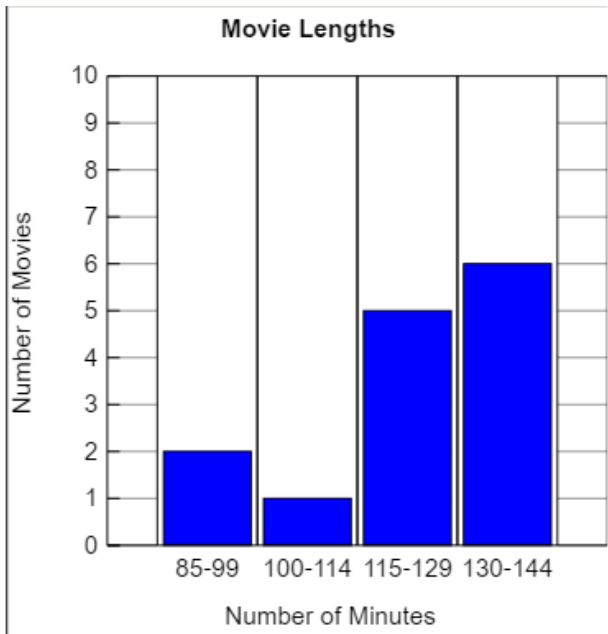
After the water was poured, $\frac{1}{4}$ gallon of water was left in the beaker.

What was the total number of **fluid ounces** of water in the beaker before the water was poured by the 12 students? (Use 1 gallon = 128 fluid ounces.) Show or explain each step you used to determine your answer.

Enter your answer and your work or explanation in the box provided.



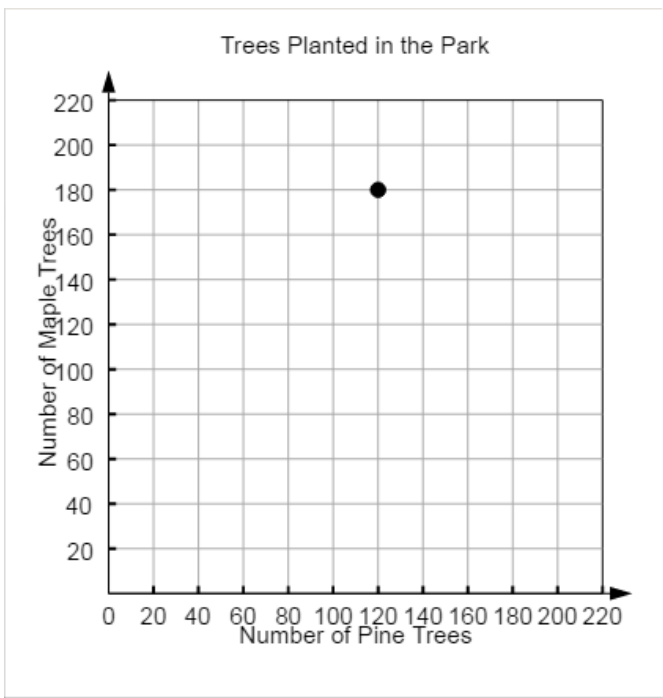
This document contains the answer keys and rubrics for the LEAP 2025 Grade 6 Mathematics Practice Test.

Session 1														
Task #	Task Type	Value (points)	Key	Alignment										
1	I	1	D	6.NS.A.1										
2	I	1	A, B, E	6.EE.A.4										
3	I	1	For every 4 mystery books checked out, 3 nonfiction books were checked out.	6.RP.A.1										
4	I	1	D	6.NS.B.3										
5	I	1	-3.5	6.NS.C.6c										
6	I	1	432	6.NS.B.2										
7	I	1	C	6.NS.C.6a										
8	I	1	B	6.NS.B.4										
9	I	1	D, E	6.SP.A.1										
10	I	1	A	6.EE.B.6										
11	I	1	Movie Lengths <table><thead><tr><th>Number of Minutes</th><th>Number of Movies</th></tr></thead><tbody><tr><td>85-99</td><td>2</td></tr><tr><td>100-114</td><td>1</td></tr><tr><td>115-129</td><td>5</td></tr><tr><td>130-144</td><td>6</td></tr></tbody></table>	Number of Minutes	Number of Movies	85-99	2	100-114	1	115-129	5	130-144	6	6.SP.B.4
Number of Minutes	Number of Movies													
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Session 1																						
Task #	Task Type	Value (points)	Key	Alignment																		
12	I	1	<table><tr><td></td><td>Above Freezing</td><td>Below Freezing</td></tr><tr><td>0.5° C</td><td>☒</td><td>☐</td></tr><tr><td>−13° C</td><td>☐</td><td>☒</td></tr><tr><td>100° C</td><td>☒</td><td>☐</td></tr><tr><td>5.5° C</td><td>☒</td><td>☐</td></tr><tr><td>−2.25° C</td><td>☐</td><td>☒</td></tr></table>		Above Freezing	Below Freezing	0.5° C	☒	☐	−13° C	☐	☒	100° C	☒	☐	5.5° C	☒	☐	−2.25° C	☐	☒	6.NS.C.5
				Above Freezing	Below Freezing																	
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			100° C	☒	☐																	
			5.5° C	☒	☐																	
−2.25° C	☐	☒																				
13	I	1	1668.2	6.NS.B.2																		
14	I	1	D	6.EE.A.1																		
15	I	1	54	6.RP.A.2																		
16	I	1	-4	6.NS.C.6c																		
17	I	1	9	6.NS.C.8																		
18	I	1	B, D	6.EE.A.4																		
19	I	1	85.104	6.NS.B.3																		
20	I	1	B	6.NS.B.3																		

Session 2				
Task #	Task Type	Value (points)	Key	Alignment
21	I	1	B	6.EE.A.2a
22	I	1	30	6.RP.A.3c
23	I	1	The ribbon costs \$0.008 per centimeter.	6.RP.A.3d
24	I	1	B	6.EE.B.5
25	I	1	164340	6.EE.A.2c
26	I	1	0.008	6.EE.A.1
27	III	3	rubric	LEAP.III.6.3 (6.RP.A.3)
28	I	2	Part A: 90 Part B: 24	6.RP.A.3c
29	I	2	Part A: A, B, E, G Part B: 16	6.G.A.3
30	II	4	rubric	LEAP.II.6.9 (5.NBT.A.1, 5.NBT.A.2)

Session 2				
Task #	Task Type	Value (points)	Key	Alignment
31	III	3	Part A: rubric Part B: rubric	LEAP.III.6.1 (6.RP.A.3b, 6.EE.A.2a, 6.EE.A.2c, 6.EE.B.6)
32	II	3	Part A: rubric Part B: rubric	LEAP.II.6.4 (6.NS.C.6a, 6.NS.C.6c)

Session 3				
Task #	Task Type	Value (points)	Key	Alignment
33	I	1	6.75	6.RP.A.3c
34	I	1		6.RP.A.3a
35	I	1	0.25	6.RP.A.3d
36	I	1	B, C	6.EE.A.2a
37	II	3	rubric	LEAP.II.6.7 (6.EE.A.4)
38	I	2	Part A: $t \times 8 = 39.60$ Part B: 4.95	6.EE.B.7
39	I	1	11	6.EE.A.2c

Session 3				
Task #	Task Type	Value (points)	Key	Alignment
40	I	2	Part A: 52 Part B: 8	6.RP.A.3b
41	II	4	rubric	LEAP.II.6.3 (6.NS.A.1)
42	I	1	B	6.EE.B.5
43	III	6	Part A: rubric Part B: rubric	LEAP.III.6.2 (5.MD.A.1, 5.MD.B.2, 5.NF.A.2, 5.NF.B.6)

RUBRICS

Task # 27	
Score	Description
3	Student response includes the following 3 elements: • Modeling component: 2 points ○ Models a strategy for developing a reasoned estimate for an appropriate length and width of each cereal bar, including explaining assumptions ○ Models a strategy for determining the amount each cereal bar will cost Megan to make • Computation component: 1 point ○ Amount each cereal bar will cost based on modeling strategy Sample Student Response: I assume that each bar could be 2 inches by 4 inches. This is a reasonable size for a cereal bar and it easy enough to hold and does not appear to be too large a serving size. The cereal bar can also be cut so that all cereal bars are the same size and shape since 24 inches and 16 inches can be evenly divided by 2 inches and 4 inches. For the 1 pan of bars cut so each bar is 2 inches by 4 inches, there would be 6 rows of bars ($24 \div 4$) and 8 bars in each row ($16 \div 2$). Altogether, that would make 48 bars for each pan. The amount spent on ingredients is \$9.85, so the amount each cereal bar will cost Megan to make is $\\$9.85 \div 48$, which is \$0.205... or about \$0.21. Notes: • Other reasoned estimates are possible. As long as the modeling steps are valid, credit should be awarded. • The student may receive a combined total of 2 points if the modeling processes are correct but the student makes one or more computational errors resulting in incorrect answers. • The student may receive a total of 1 point if he/she computes the correct answer but shows no work or insufficient work to indicate a correct modeling process.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #30

Score	Description
4	Student response includes the following 4 elements: • Reasoning component: 2 points ○ Correctly explains why Pattern A is incorrect ○ Correctly explains why Pattern B is incorrect • Computation component: 2 points ○ Correct values for Pattern A ○ Correct values for Pattern B Sample Student Response: The student added zeros to the right of the number, instead of moving the number up one place value. The student added zeros to the left of the decimal portion of the number, instead of moving the number down one place value. For pattern A $3.675 \times 10 = 36.75$ $3.675 \times 100 = 367.5$ $3.675 \times 1,000 = 3,675$ For Pattern B $3.675 \times 0.1 = 0.3675$ $3.675 \times 0.01 = 0.03675$ $3.675 \times 0.001 = 0.003675$ Note: Other valid reasoning exists. As long as the student explains the flaw in the provided work, credit should be awarded.
3	Student response includes 3 of the 4 elements.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #31	
Part A	
Score	Description
1	Student response includes the following element: • Modeling component: 1 point ○ Correct expression that represents the total amount of money raised Sample Student Response: 15x Note: Any valid equivalent expression can receive credit.
0	Student response is incorrect or irrelevant.
Part B	
Score	Description
2	Student response includes the following 2 elements: • Modeling component: 1 point ○ Shows or explains a correct process to find the difference • Computation component: 1 point ○ Correct answer, 145 Sample Student Response: $15 \times 43 = 645$, and $645 - 500 = 145$ OR Using my expression, I multiplied 43 by \$115 to get a total of \$645 raised. I then subtracted \$500 from \$645 to get \$145 for the amount that the club exceeded its goal. Notes: • The student may receive 1 point for Part B if the modeling process is correct but the student makes one or more computational errors resulting in incorrect answers. • The student may receive 1 point for Part B if he or she computes the correct answers but shows no work or insufficient work to indicate a correct modeling process. • If a student writes an incorrect model and answers the remaining prompts based on the model, he or she can receive 1 point for computation but no points for modeling.
1	Student response includes 1 of the 2 elements.
0	Student response is incorrect or irrelevant.

Task #32	
Part A	
Score	Description
2	Student response includes the following 2 elements: • Reasoning component: 1 point ○ Correct work shown or explanation given using the number line • Computation component: 1 point ○ Correct distance of each point from Q (0.3 for R and 0.6 for P) Sample Student Response: Point R is 0.3 unit from point Q, because there are 3 spaces of 0.1 between them on the number line. Point P is 0.6 unit from point Q, because there are 6 spaces of 0.1 between them on the number line.
1	Student response includes 1 of the 2 elements.
0	Student response is incorrect or irrelevant.
Part B	
Score	Description
1	Student response includes the following element: • Reasoning component: 1 point ○ Correct explanation of how to find point S on the number line Sample Student Response: Since point Q is at 0 and since point S is the same distance from point Q as point R but in a different location, it must be on the opposite side of point Q. Points R and S are on opposite sides of 0 on the number line, so their locations should have opposite signs. Since point R is located at 0.3, point S must be located at -0.3. Note: Point S can also be located at 0.3 for credit with a valid explanation.
0	Student response is incorrect or irrelevant.

Task #37

Score	Description
3	Student response includes the following 3 elements: • Reasoning component: 2 points ○ Correct explanation of why Brianna's thinking is incorrect ○ Correct explanation of how to determine which expressions are equivalent • Computation component: 1 point ○ Identifies expressions A and C as equivalent Sample Student Response: Brianna only checked the value of each expression for one substitution of x. To check which expressions are equivalent, I need to check that they are the same value for any substitution of x. Since expressions A and C are not equivalent to the expression $6x - 4$, they will be equivalent for any substitution of x, so they are equivalent.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #41

Score	Description
4	Student response includes the following 4 elements: • Reasoning component: 3 points ○ Correct explanation of how to find the number of sheets in a stack using the ruler ○ Correct expression or equation that can be used to find the number of sheets, $2\frac{1}{4} \div \frac{3}{16}$ or equivalent ○ Correct explanation of how expression relates to use of the ruler • Computation component: 1 point ○ Correct number of sheets of cardboard in a stack, 12 Sample Student Response: To find the number of sheets in a stack using the ruler, you start at $2\frac{1}{4}$ inches on the ruler. Then you can mark off groups of $\frac{3}{16}$. This is 3 of the 16ths marks on the ruler. Then you can count the number of groups. There were 12 groups, so there are 12 sheets in a stack. An expression that represents this is $2\frac{1}{4} \div \frac{3}{16}$. This relates to using the ruler because you are starting with $2\frac{1}{4}$ and dividing by $\frac{3}{16}$, which is really finding how many groups of $\frac{3}{16}$ there are in $2\frac{1}{4}$. When you divide, you will get 12, which means there are 12 groups of $\frac{3}{16}$ in $2\frac{1}{4}$.
3	Student response includes 3 of the 4 elements.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #43

Part A

Score	Description
3	Student response includes the following 3 elements: • Computation component: 1 point ○ Correct total number of cups of water, 3 • Modeling component: 2 points ○ Correct expression using addition AND multiplication ○ Correct process for evaluating the expression written Sample Student Response: 3 (cups) $6 \times \frac{1}{8} + 2 \times \frac{1}{4} + 3 \times \frac{3}{8} + 1 \times \frac{5}{8}$ $6 \times \frac{1}{8} + 2 \times \frac{1}{4} + 3 \times \frac{3}{8} + 1 \times \frac{5}{8} =$ $\frac{6}{8} + \frac{2}{4} + \frac{9}{8} + \frac{5}{8} =$ $\frac{6}{8} + \frac{4}{8} + \frac{9}{8} + \frac{5}{8} = \frac{24}{8} = 3$ Notes: • The student must show operations of addition AND multiplication in order to receive the modeling point. If students only use addition, they do not get the modeling point. • The student must show only one expression to receive this modeling point. • If the student writes an incorrect expression but shows a correct process for evaluating that expression, the student will receive 1 modeling point.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.

Task #43

Part B

Score	Description
3	Student response includes the following 3 elements: • Computation component: 1 point ○ Correct total number of fluid ounces, 56 fluid ounces • Modeling component: 2 points ○ Correct process for finding the amount of water in the beaker ○ Correct process for converting gallons and cups to fluid ounces Sample Student Response: The amount of water in the beaker can be found by adding 3 cups to $\frac{1}{4}$ gallon. To convert $\frac{1}{4}$ gallon to fluid ounces, I need to multiply by 128, which is 32 fluid ounces. To convert 3 cups to fluid ounces, I need to multiply by 8, which is 24 fluid ounces. The amount of water in the beaker before the water was poured out is $32 + 24 = 56$ fluid ounces. Notes: • Units are not required to receive credit. • The student may receive a combined total of 4 points if the modeling processes are correct but the student makes one or more computational errors resulting in incorrect answers. • The student may receive a total of 2 points if he or she computes the correct answers but shows no work or insufficient work to indicate a correct modeling process. • The student cannot receive more than 2 points for modeling if the explanations, while sufficient to indicate that the student had a correct process contain nonsense statements, such as $\frac{1}{4} \times 128 = 32 + 24 = 56$.
2	Student response includes 2 of the 3 elements.
1	Student response includes 1 of the 3 elements.
0	Student response is incorrect or irrelevant.